SPRING

2026

April *Skies*

NORTHERN HEMISPHERE

clearskies.starstucklab.com

"The cosmos is also within us — we are a way for the universe to know itself."

— Carl Sagan

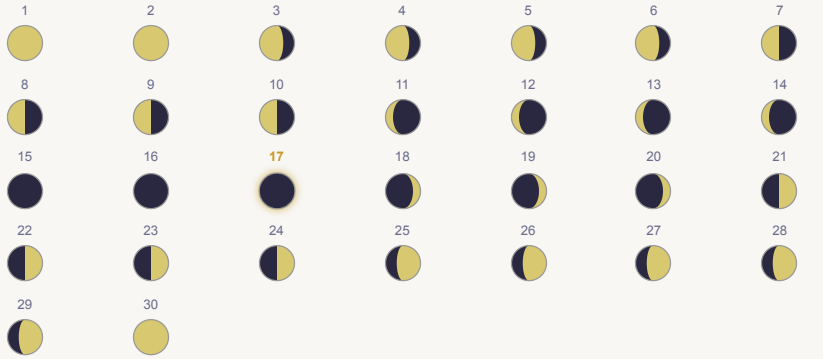
Sky Planner — April 2026

Full: Apr 2 · Last Qtr: Apr 10 · New: Apr 17 · First Qtr: Apr 24

DATES	BEST TARGETS	BEST TIME	NOTE
Apr 1–3	Moon, Jupiter, Venus (low W), M42 & M46/47 (low SW)	8–10pm	Full Moon — skip faint deep sky
Apr 1–30	Venus low WNW, climbing higher each week	8:30–9:30pm	Clear WNW horizon essential
Apr 1–12	M42 Orion Nebula, M46 & M47 separately	8:30–10pm	Window closing — sets ~11pm
Apr 4–9	Moon craters, Algieba double star	10pm–1am	Moon sets ~2am, improving each night
Apr 10–16	M44, M3, Algieba, Leo Triplet	10pm–2am	Good dark window after midnight
Apr 17–20	★ All spring objects — best of month	9pm–3am	New Moon Apr 17 — darkest skies
Apr 21–22	★ Lyrid meteors + all deep sky	10pm–4am	Lyrid peak — 10–15/hr, Moon down
Apr 10–20	Comet C/2025 R3 pre-dawn (mag ~8)	90min before sunrise	Enters solar glare after Apr 20
Apr 28–30	Moon craters (waxing gibbous)	8–11pm	Moon brightening — deep sky fading

Venus → low WNW at dusk, climbing all month · Orion/Puppis → low SW 8:30pm · Leo/Virgo → due south 11pm · M44 → south-SW · M3 → east-SE by midnight

The Moon — April 2026



Full — Apr 2 · Last Qtr — Apr 10 · New Moon — Apr 17 · First Qtr — Apr 24

Clavius crater

One of the largest craters on the Moon. At 100× look for smaller craters inside it along the southern terminator — a crater within a crater.

Best: Apr 19–21 · 9mm (100×)

Apennine mountains

Dramatic range bordering Mare Imbrium. Peaks cast sharp black shadows at low illumination near the terminator — spectacular at 36×.

Best: Apr 20–22 · 25mm (36×)

Tycho crater

Young, sharp-walled crater in the southern highlands. Ray system visible at full moon; walls most dramatic Apr 19–20.

Best: Apr 19–20 · 9mm (100×)

Earthshine

Ghostly glow on the dark limb of the crescent Moon. Naked eye or binoculars only — the telescope makes it too bright to appreciate.

Apr 13–14 pre-dawn · Apr 20–21 post-sunset

Planets & Events — April 2026

Jupiter ★★★★★

Constellation	Gemini
Visible	Evening, SW sky
Sets after sunset	2–3 hrs
Brightness	mag −2.1
At 36× (25mm)	Disc + 4 Galilean moons
At 100× (9mm)	Cloud bands, equatorial belts

Galilean moons change configuration nightly. Crescent Moon nearby Apr 22–23.

Venus ★★★★☆

Constellation	Aries → Taurus (from Apr 19)
Visible	Evening, low western horizon
Sets after sunset	~1.5 hrs (Apr 1) → ~2.5 hrs (Apr 30)
Brightness	mag −3.9 — blazing white
At 36× (25mm)	Tiny gibbous disc, ~11–12″
At 100× (9mm)	94→88% lit, featureless disc

⚠ Low horizon essential — best from an elevated spot facing WNW. Crescent Moon nearby Apr 18–19 near the Pleiades. Passes close to Uranus Apr 23–24.

Venus — brilliant but low, clear horizon required

Venus blazes at mag −3.9 in the west-northwest after sunset all month — the brightest object in the sky after the Moon. From 32°N it sits only ~13° high 30 min after sunset early in the month, climbing to ~20° by month's end. An unobstructed WNW horizon is essential. Through the telescope: a small, nearly full gibbous disc (11–12″, 88–94% li…

Mars — pre-dawn only, very difficult in April 2026

Mars is a morning planet rising low in the east before dawn. Extremely difficult from northern latitudes this month due to low altitude. Skip until autumn 2026.

Comet C/2025 R3 — pre-dawn window Apr 10–20 only

Magnitude ~8 comet in Pegasus/Pisces, visible at 36× (25mm) ~90 min before sunrise, low east. Enters solar glare around Apr 20–25. Check sky reports before going out.

+ 2 more events — see full guide online

M42 — Orion Nebula

Emission nebula · Orion · mag 4.0 · 1,350 ly

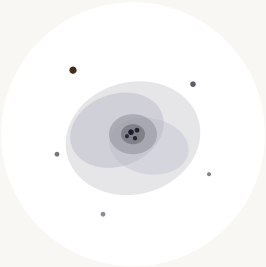
★ ★ ★ ★ ★ OBSERVING EASE

⚠ April 2026 window closing fast — Orion is setting. Observe 8:30–10pm in the first two weeks only, low southwest. Gone from useful observing by mid-April.

BEST DATES	BEST TIME	DIRECTION
Apr 1–12	8:30–10pm only	Low SW, sinking
START WITH	**THEN TRY**	**MOON CAUTION**
25mm (36×)	9mm (100×)	Survives most phases

DESCRIPTION

At 36× an unmistakable fan-shaped glow — two bright lobes spread like wings around a luminous core. At 100× the Trapezium becomes the centrepiece: four hot young stars in a tiny trapezoid at the nebula's heart, each cleanly split as individual pinpoints. The surrounding nebulosity gains texture — brighter regions, darker bays. One of the most spectacular objects any telescope can show.



25mm · 36×
fan glow + Trapezium

FINDER

Find Orion's Belt — three stars in a row, low SW. Below the belt hangs the Sword — a short chain of fainter points. The middle "star" of the Sword looks fuzzy: that's M42. Completely unmistakable.

M46 & M47 — Twin Clusters

Open clusters · Puppis · mag 6.1 (M46) / 4.2 (M47) · observe separately

★★★★☆ OBSERVING EASE

⚠ Early April 2026 only — low SW, sinking fast. Observe as soon as it's fully dark. Not usefully visible after mid-April from northern latitudes.

BEST DATES	BEST TIME	DIRECTION
Apr 1–10	8:30–10:30pm	Low SW, Puppis
USE	**SEPARATION**	**BONUS IN M46**
25mm (36×) each	~1.3° — outside FOV	NGC 2438 nebula

DESCRIPTION

M46 and M47 are 1.3° apart — just outside your 1.44° true FOV at 36×. Observe each separately, then sweep between them to appreciate the contrast. M47 is the showpiece: a bright loose scatter of prominent stars with warm yellow-orange giants. M46 is a complete contrast — a much denser, uniform cluster of fainter stars, like a fine dusting of salt. At 100× look on M46's northern edge for NGC 2438: a tiny, ghostly ring — a planetary nebula in the foreground of the cluster.

M47 · 25mm · 36×
bright & loose

M46 · 25mm · 36×
dense · NGC 2438 ring

FINDER

Find Sirius — the brightest star, blazing SW. Scan ~12–14° east and slightly north (~1.5 fist-widths). M47 appears first as a bright patch; M46 is ~1.3° further east. Sweep slowly between them.

M44 — Beehive Cluster

Open cluster · Cancer · mag 3.7 · 577 ly

★ ★ ★ ★ ★ OBSERVING EASE

BEST DATES	BEST TIME	DIRECTION
Apr 4–28	9pm–1am	South-SW, high
USE ONLY	**MOON CAUTION**	**SKIP**
25mm (36×)	Survives most phases	Apr 1–3 (full moon)

DESCRIPTION

A spectacular loose swarm of stars filling the whole field. M44 spans ~1.6° — slightly wider than your 1.44° FOV — so stars drift off the edge giving a sense of immersion. Don't switch to the 9mm. Look for the subtle warm yellowish tinge of the K-type giant stars. Naked-eye as a fuzzy patch on a clear night.

25mm · 36×
loose star swarm, fills FOV

FINDER

Find Castor and Pollux in Gemini (bright pair high SW). Move two fist-widths east — fuzzy naked-eye patch on a clear night. That's M44. Centre it in the 25mm.

M3 — Globular Cluster

Globular cluster · Canes Venatici · mag 6.2 · 34,000 ly

★★★★☆ OBSERVING EASE

BEST DATES	BEST TIME	DIRECTION
Apr 10–27	11pm–2am	East, rising high
START WITH	THEN TRY	NEEDS
25mm (36×)	9mm (100×)	Dark skies

DESCRIPTION

At 36× an obvious round cotton-ball glow with a concentrated bright core. At 100× the halo takes on a slightly granular, salt-and-pepper texture at the edges — tantalising hints of stars, but not cleanly resolved pinpoints. A 114mm aperture cannot fully resolve M3's stars — that requires 150mm or more. The granular texture is real and rewarding. Use averted vision to draw it out.

9mm · 100×
granular halo, bright core

FINDER

Find bright orange Arcturus high east by midnight. Move ~6° northwest toward Cor Caroli in Canes Venatici, then nudge south. M3 appears as a fuzzy "star" that refuses to sharpen with focus.

Algieba — γ Leonis

Binary star · Leo · combined mag 2.0 · 126 ly · separation ~4.6″

★★★★☆ OBSERVING EASE

BEST DATES	BEST TIME	DIRECTION
Apr 4–27	9pm–1am	South, high in Leo
USE	**NEEDS**	**MOON CAUTION**
9mm (100×)	Steady seeing	Any phase fine

DESCRIPTION

At 100× two warm golden-orange stars side by side, cleanly separated — one of the most beautiful colour-matched pairs in the sky. Both are giant stars with nearly identical warm hues. Separation ~4.6 arcseconds — comfortably within reach of 114mm on a steady night. If the air is unsteady the pair blurs into one — try again on a calmer night.

9mm · 100×
golden pair, cleanly split

FINDER

Find the Sickle of Leo — the backward question-mark with Regulus at the base. Algieba is the bright star near the top of the curve, third brightest in Leo. Naked-eye easy. Point the 9mm at it — steady air needed to split cleanly.

Leo Triplet — M65, M66, NGC 3628

Galaxy group · Leo · mag 9.3–10.2 · 35 million ly

★★☆☆☆ OBSERVING EASE

BEST DATES	BEST TIME	DIRECTION
Apr 10–25	10pm–1am	Due south, high
USE	**NEEDS**	**AVOID**
25mm (36×)	Bortle 4 or better	Any moonlight

DESCRIPTION

Genuinely challenging for 114mm. On a truly dark night (Bortle 4+), M65 and M66 appear as faint elongated oval smudges with slightly brighter cores. NGC 3628 (the "Hamburger galaxy") is significantly fainter — a thin, low-contrast edge-on streak requiring averted vision and excellent transparency. Completely invisible from light-polluted skies. Reserve for the Apr 17–20 new moon window only.

25mm · 36×
faint ovals, dark skies only

FINDER

Find Theta Leonis (θ Leo) — bottom-right star in Leo's body. M65 and M66 sit ~1.5° south (three moon-widths). Sweep slowly south at 36×. NGC 3628 is ~0.6° further north — averted vision and patience required.

M51 — Whirlpool Galaxy

Spiral galaxy + NGC 5195 · Canes Venatici · mag 8.4 · 23 million ly

★★★☆☆ OBSERVING EASE

BEST DATES	BEST TIME	DIRECTION
Apr 10–27	11pm–2am	North, high overhead
START WITH	**THEN TRY**	**NEEDS**
25mm (36×)	9mm (100×)	Excellent dark skies

DESCRIPTION

At 36× two fuzzy glows close together — M51 and companion NGC 5195. At 100× M51's core brightens and both nuclei are distinct. The faint bridge may be hinted at with averted vision on the very best nights. Note: spiral arms are NOT visible at 114mm — that requires 150–200mm. What you see is a round, uneven glow. Still the showpiece of the month.

9mm · 100×
two nuclei, hint of bridge

FINDER

Find Alkaid — end star of the Big Dipper's handle. M51 sits ~3.5° southwest (7 moon-widths). At 36× sweep slowly — the double glow is the giveaway.

Observing Conditions

Seeing (steadiness)

Poor seeing makes stars shimmer and prevents double-star splits. Good seeing gives steady, sharp images. Seeing varies nightly — try a different night if stars are boiling.

Transparency (clarity)

High transparency means dry, dust-free air. Low transparency hides faint galaxies. Haze and humidity are the enemy of faint objects.

Dark adaptation

Your eyes need ~20 minutes in full darkness to reach peak sensitivity. One bright light flash resets the process. Use red light only in the field.

Light pollution

Bortle 1–4 skies reveal galaxy detail and faint nebulosity. The Moon is the single biggest factor night-to-night.

Pre-session checklist

- [ ] Check seeing/transparency forecast (Clear Outside · Meteoblue)
- [ ] Dark sky window: **Apr 10–27**
- [ ] Cool telescope ≥20 min before observing
- [ ] Collimate if last session was ≥2 weeks ago
- [ ] Always start at 25mm (36×) — low-power first
- [ ] Switch to 9mm (100×) only after locating target
- [ ] 20 min dark adaptation before faint objects
- [ ] Use averted vision for galaxies and faint nebulae

Glossary

Magnitude (mag)

Brightness scale — lower is brighter. Sun −26, naked eye limit ~6, your scope reaches ~12.

True FOV

AFOV ÷ magnification. At 36× with 52° AFOV: 52÷36 = 1.44°. The Moon is 0.5° wide.

Averted vision

Look slightly off-centre to use rod cells. Essential for faint galaxies and M3's halo texture.

Seeing

Atmospheric steadiness. Poor seeing blurs stars and prevents double star splits.

Terminator

Light/dark boundary on the Moon — the best place to see shadow-enhanced crater detail.

Dark adaptation

Eyes need ~20 min in full darkness for peak sensitivity. One white light flash resets it.

Arcsecond (″)

1/3600 of a degree. Algieba's 4.6″ gap is comfortably split at 100× on a steady night.

Bortle scale

1–9 sky darkness scale. Leo Triplet and NGC 3628 need Bortle 4 or better.

Trapezium

Four hot young stars at M42's core — clearly split as pinpoints at 100×.

AFOV

Apparent field of view of the eyepiece. Your 25mm has ~52°. Divide by magnification for true FOV.

Your 114mm f/7.9 reflector's limits: limiting magnitude ~12 under dark skies · max useful magnification ~230× (100× practical ceiling most nights) · 265× more light than the naked eye · resolving power ~1 arcsecond on the best nights · cannot fully resolve globular clusters — only granular halo texture at the edges.

CLEAR SKIES — APRIL 2026

Anish Mangal

With thanks to: Ishan Chrungoo, Sohail Lalani

114mm f/7.9 Newtonian · 900mm focal length · 25mm (36×) & 9mm (100×)

"The cosmos is also within us — we are a way for the universe to know itself."

— Carl Sagan

My Notes